
Learning to Rotate: Temporal and Semantic Rotary Encoding for Sequential Modeling

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Abstract

Every Transformer architecture dedicates enormous capacity to learning rich representations in *semantic embedding space*—yet the rotation manifold acted upon by Rotary Positional Embeddings (RoPE) has been treated as a fixed, hand-crafted structure, populated only by discrete ordinal indices. We argue that this *rotation space* is a largely overlooked second dimension of expressivity in the attention mechanism, one whose systematic exploration may open a new door for attention-based architectures. The analogy to complex numbers is instructive: just as introducing the imaginary axis—orthogonal to and independent of the real line—unlocked new algebraic structure once believed impossible, treating the rotation manifold as a learnable, signal-conditioned space opens an orthogonal degree of freedom in attention. In this framing, the token embedding encodes the *semantic* (real) component of a representation—*what* a token means—while the rotation encodes its *dynamic* (imaginary) component—*how* it relates to every other token across time, position, and context.

We introduce **SIREN-RoPE**, a concrete instantiation of this idea, which populates the rotation dimension with heterogeneous signals—continuous timestamps, cyclical temporal patterns, and categorical metadata—via a dual-branch Sinusoidal Representation Network (SIREN) [Sitzmann et al., 2020]. As a proof of concept, we evaluate on a production-scale news feed dataset from a major social network using a generative recommender as the ranking model, demonstrating that activating this hidden dimension yields consistent improvements across calibration and ranking objectives with negligible computational overhead. We invite the community to view the rotation space not as a solved positional-encoding detail, but as an untapped axis whose rich structure may prove as consequential for attention as the imaginary unit proved for algebra. Demo code is available at https://github.com/hailingc/siren_rope.

1 Introduction

Rotary Position Embeddings (RoPE) [Su et al., 2024] in Large Language Models (LLMs) encode sequence order as planar rotations of query/key subspaces, yielding strong relative-position modeling. Yet the rotation manifold itself has always been a fixed, hand-crafted structure driven by discrete ordinal indices. In sequential recommendation and event-stream modeling, this is a poor proxy: an interaction seven days ago is qualitatively different from one seven minutes ago, even at the same ordinal offset.

We argue this points to a deeper opportunity. Just as extending the real line to the complex plane—by adding an orthogonal imaginary axis—unlocked entirely new algebraic structure, treating the RoPE rotation manifold as a learnable, signal-conditioned space opens an orthogonal expressive dimension

in attention. Token embeddings encode the *semantic* (real) component—*what* a token means; the rotation encodes the *dynamic* (imaginary) component—*how* it relates to every other token across time and context. This dimension has been almost entirely unexplored.

1.1 Contributions

We introduce **SIREN-RoPE**, which treats the rotation manifold as a first-class learnable space:

- **Temporal rotation.** A dual-branch SIREN–DNN network maps continuous timestamps into per-dimension rotation angles, capturing both periodic (daily/weekly) and aperiodic (recency decay) temporal structure directly in the geometry of attention.
- **Adaptive frequency learning.** Per-dimension frequency scales and an ordinal-gate scalar λ are jointly learned, replacing hand-crafted inverse-frequency constants.
- **Empirical validation.** On a production-scale social feed dataset, SIREN-RoPE consistently improves calibration (NE) and ranking (AUC) across three engagement tasks with $\sim 0.2\%$ extra parameters. Ablations confirm the rotation dimension carries signal complementary to and independent of the embedding space.

2 Related Work

Positional Encodings and RoPE. Early Transformers inject order via fixed sinusoidal or learned absolute embeddings [Vaswani et al., 2017]; subsequent work shifted to relative formulations for better length generalization [Shaw et al., 2018, Dai et al., 2019, Raffel et al., 2020]. RoPE encodes position as planar rotations in query-key subspaces, expressing relative offsets as phase differences while preserving vector norms [Su et al., 2024]. Extensions such as YaRN and LongRoPE focus on length extrapolation via interpolation [Peng et al., 2023, Ding et al., 2024]. All of these treat the rotation manifold as a fixed positional scaffold; we instead treat it as a learnable, signal-conditioned space.

Temporal Dynamics in Sequential Models. Methods like Time2Vec [Kazemi et al., 2019] and TiSASRec [Li et al., 2020] incorporate pairwise time intervals and periodic embeddings to capture non-linear temporal dynamics. DeBERTa [He et al., 2021] and TUPE [Ke et al., 2021] disentangle content from positional context in language modeling. Most closely related to our work, TO-RoPE [Wei et al., 2025] applies the same inverse-frequency schedule as standard RoPE but substitutes the raw timestamp T_i for the ordinal index, yielding a per-dimension rotation angle $T_i \cdot \theta_j$. SIREN-RoPE generalizes this idea by replacing the fixed inverse-frequency mapping with a dual-branch SIREN–DNN network that maps rich multi-scale temporal features into learned rotation angles, enabling the model to capture complex periodicities that the fixed frequency schedule cannot express.

Implicit Neural Representations and Continuous-Time Models. SIRENs [Sitzmann et al., 2020] use periodic activations with principled initialization to overcome spectral bias [Tancik et al., 2020], enabling multi-scale periodic function learning. INR-based methods have been adapted for time-series [Fons et al., 2022, Le Naour et al., 2024] but not for attention geometry. Neural ODEs/CDEs [Chen et al., 2018, Kidger et al., 2020] and hybrid models like ContiFormer [Chen et al., 2024] handle continuous-time sequences at significant inference cost. SIREN-RoPE injects continuous-time inductive bias directly into the rotary angle, achieving comparable expressivity at the cost of a small MLP.

3 SIREN-RoPE: Unified Rotary Encoding

3.1 Problem Formulation

Let $\mathcal{S} = \{(\mathbf{e}_i, T_i)\}_{i=1}^C$ denote a user interaction sequence of length C , where $\mathbf{e}_i \in \mathbb{R}^d$ is the item embedding and $T_i \in \mathbb{R}$ represents the interaction timestamp. Conventional Rotary Positional Embedding (RoPE) [Su et al., 2024] modulates the inner product between query and key representations using rotations derived from discrete ordinal indices p_i . While the attention weights are primarily driven by the semantic content of $\mathbf{e}_i$, this formulation constrains the relative positional dependency to

be a strictly periodic function of the ordinal displacement $|p_i - p_j|$. Consequently, it neglects the irregular temporal intervals $\Delta T = T_i - T_j$ that often characterize real-world user behavior.

In social-feed recommendation, treating ordinal position as a proxy for time is fundamentally misaligned with empirical user behavior. An interaction at 08:00 on a Monday is qualitatively distinct from one at 22:00 on a Saturday, even when they share the same ordinal displacement $|p_i - p_j|$. Human behavior is governed by strong periodicities—**24-hour diurnal cycles** and **7-day weekly rhythms**—so user intent and content relevance are tied to absolute temporal coordinates, a structure that ordinal indices alone cannot capture. We therefore seek a rotation angle $\Theta_j(T_i, p_i)$ satisfying three criteria:

1. **Temporal richness:** captures multi-scale cyclical patterns from T_i .
2. **Ordinal preservation:** retains the monotone recency-decay and translational equivariance properties of standard RoPE.
3. **Adaptive balance:** allows end-to-end gradient descent to determine how much each component contributes.

3.2 Ordinal-Temporal Fusion

SIREN-RoPE replaces the fixed angle $p_i \theta_j$ with:

$$\Theta_j(T_i, p_i) = \underbrace{f_\phi(T_i)_j \cdot \omega_j^s}_{\text{Temporal (SIREN)}} + \underbrace{p_i \cdot \theta_j \cdot \lambda}_{\text{Ordinal (scaled)}} \quad (1)$$

where:

- $i \in \{0, \dots, C-1\}$ is the item’s ordinal index in the sequence;
- $p_i = i$ is the ordinal position of item i ;
- $f_\phi : \mathbb{R}^{d_t} \rightarrow \mathbb{R}^{d_k/2}$ is the dual-branch SIREN network (Section 3.3), mapping d_t -dimensional timestamp features to rotation angles;
- $\omega_j^s \in \mathbb{R}^{d_k/2}$ are learnable per-dimension frequency scalings;
- $\theta_j = \text{base}^{-2j/d_k}$ are the fixed inverse frequencies used in standard RoPE;
- $\lambda \in \mathbb{R}$ is a learnable scalar gate (initialized to 1.0) controlling the ordinal contribution.

The rotation is then applied identically to standard RoPE with Θ_j substituted for $p\theta_j$, where $\mathbf{q}$ is the query vector and $\mathbf{k}$ is the key vector:

$$q'_{2i} = q_{2i} \cos(\Theta_j) - q_{2i+1} \sin(\Theta_j), \quad (2)$$

$$q'_{2i+1} = q_{2i} \sin(\Theta_j) + q_{2i+1} \cos(\Theta_j). \quad (3)$$

The same rotation applies to $\mathbf{k}$. This real-number formulation avoids complex-number operations and is fully compatible with `torch.compile`.

Initialization and convergence. At initialization ($\omega_j^s = \pi, \lambda = 1.0$), the model begins close to a scaled form of standard RoPE augmented by a SIREN offset, providing a stable starting point. As training progresses, ω_j^s and λ learn to balance temporal and ordinal contributions. In practice, we observe a striking pattern: without temporal signals (ordinal RoPE only), λ remains near its initial value of 1.0 throughout training. Upon introducing SIREN-RoPE, however, λ drops to 0.044 at convergence, indicating that the model has learned to rely predominantly on temporal modulation over ordinal position—the rotation dimension, once given a rich temporal signal to carry, largely supplants the need for discrete ordinal recency.

3.3 Dual-Branch SIREN Architecture

f_ϕ additively combines a periodic SIREN branch and an aperiodic DNN branch:

$$f_\phi(T) = \underbrace{f_{\sin}(T)}_{\text{Periodic branch}} + \underbrace{f_{\text{DNN}}(T)}_{\text{Aperiodic branch}}. \quad (4)$$

Periodic branch $f_{\sin}$ uses the SIREN architecture [Sitzmann et al., 2020], where each hidden layer computes $\sin(\omega_0 \mathbf{W}\mathbf{x} + \mathbf{b})$. The sine activations enable the branch to autonomously discover temporal periodicities beyond the manually specified daily/weekly cycles.

Aperiodic branch f_{DNN} uses standard ReLU activations to capture monotone trends such as content-recency decay. When the input features are themselves cyclical (as with the $(\cos, \sin)$ pairs in Eq. (5)), the DNN branch retains the flexibility to model periodic components as well.

Full implementation details are available at https://github.com/hailingc/siren_rope.

3.4 Temporal Input Features

The input to f_ϕ is a 5-dimensional decomposition of the raw Unix timestamp T :

$$\mathbf{t}(T) = \left[\cos\left(\frac{2\pi T}{\tau_d}\right), \sin\left(\frac{2\pi T}{\tau_d}\right), \cos\left(\frac{2\pi T}{\tau_w}\right), \sin\left(\frac{2\pi T}{\tau_w}\right), \tilde{T} \right], \quad (5)$$

where $\tau_d = 86,400$ s (daily periodicity), $\tau_w = 604,800$ s (weekly periodicity), and $\tilde{T}$ is a normalized long-range offset. Representing each cycle as a $(\cos, \sin)$ pair ensures continuity across period boundaries (midnight, end of week) and avoids phase-discontinuity artifacts. Notably, in this rotational representation the model need not know the user’s timezone to capture morning or evening patterns; the periodicity is recovered automatically from the behavioral sequence.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on a production-scale dataset collected from the news feed of a major social network, comprising user–item interaction events from one year of production logs. Each user’s history is represented as an interleaved sequence $[\mathbf{e}_1, \mathbf{a}_1, \mathbf{e}_2, \mathbf{a}_2, \dots, \mathbf{e}_C, \mathbf{a}_C]$, where $\mathbf{e}_i$ is the item representation and $\mathbf{a}_i$ is the corresponding action representation; the action labels serve as prediction targets. Full details of this input format are described in Cheng [2026]. Each event carries a Unix timestamp, user and item features, and engagement labels across three tasks: **Contribution** (like, comment or share actions), **Like**, and **LongDwell** (sustained dwell-time engagement). Temporal patterns are pronounced due to weekday/weekend and morning/evening behavioral cycles. The dataset reflects production-level complexity in terms of interaction volume, item diversity, and behavioral heterogeneity, representing conditions substantially more demanding than standard academic benchmarks. Our goal is not to establish a new benchmark comparison, but to demonstrate on a real production system that the rotation dimension carries predictive signal worth the community’s attention.

Baselines. All four models (three baselines and SIREN-RoPE) share the identical AttnMVP backbone, feature set, and training procedure. The *only* controlled variable is how—or whether—the item timestamp is used:

- **Ordinal RoPE:** Standard RoPE with discrete ordinal position indices. *No timestamp is used anywhere in the model.* This is the production baseline.
- **Timestamp-as-Feature:** Standard ordinal RoPE, but the 5-dimensional timestamp features $\mathbf{t}(T)$ (Eq. (5)) are appended to the item sequence features before the Transformer stack. The rotation angle remains purely ordinal. This baseline isolates whether the timestamp carries useful signal when routed through the *embedding* (semantic) dimension rather than the rotation dimension.
- **TO-RoPE** (Time and Order RoPE) [Wei et al., 2025]: Applies the same inverse-frequency schedule as standard RoPE ($\theta_j = \text{base}^{-2j/d_k}$) but uses the normalized timestamp $\tilde{T}_i$ as the index instead of the ordinal position, yielding rotation angle $\tilde{T}_i \cdot \theta_j$ per dimension. No timestamp is added to the sequence features. This baseline tests whether adding static time frequencies in the standard RoPE formula suffices.

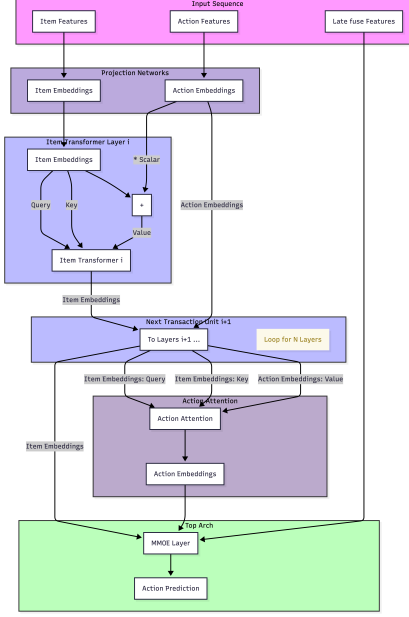


Figure 1: Base model architecture (AttnMVP, Cheng [2026]). Item and action embeddings are maintained as separate streams. Item embeddings form the Query/Key projections in every Transformer layer while action embeddings are mixed into the Value stream under strict causal masking. A final action-pooling step aggregates historical action signals conditioned on item-level similarity before the prediction head.

Base Model Architecture. All experiments use the **AttnMVP** (Attention-based Mixed Value Pooling) backbone proposed by Cheng [2026]; Figure 1 provides an overview. AttnMVP maintains item and action embeddings as separate streams, departing from the interleaved token formulation of prior generative recommenders [Zhai et al., 2024]. Within each Transformer layer, item embeddings serve as both Queries and Keys, while action embeddings are mixed additively into the value stream: $\mathbf{V}_t = \mathbf{H}_t + \alpha \mathbf{a}_t$, where $\mathbf{H}_t$ is the current item representation and α controls the action contribution. Strict causality is enforced via a *query-shifting* mechanism compatible with FlashAttention: queries are shifted one step left relative to the keys, so the representation of item i_n attends only to positions $\{0, \dots, n-1\}$, preventing target-label leakage through the value stream. After the final Transformer layer, a causally masked action-pooling step aggregates historical action embeddings conditioned on item similarity, producing the action-aware vector fed to the prediction head.

Every model variant in this paper uses this identical backbone with the identical item, action, and context feature set. The sole controlled variable across models is how the timestamp is incorporated—Table 1 summarizes the design space. For SIREN-RoPE and TO-RoPE the timestamp enters *only* through the rotation angle and is never added to the sequence features. For the Timestamp-as-Feature baseline the timestamp enters *only* through the sequence features and the rotation angle remains standard ordinal RoPE. Ordinal RoPE uses no timestamp anywhere.

Table 1: How each model incorporates the item timestamp. All other components (backbone, features, training) are identical.

Model	Timestamp in sequence features	Timestamp in rotation angle
Ordinal RoPE	×	×
Timestamp-as-Feature	✓	×
TO-RoPE	×	time-domain inverse frequencies
SIREN-RoPE (ours)	×	dual-branch SIREN

Implementation details. All models share the production Transformer backbone: 12 Transformer layers, $d = 512$ hidden dimensions, 4 attention heads, maximum sequence length $C = 1024$. We

Table 2: Engagement prediction on the production social feed dataset. NE: lower is better; AUC: higher is better. Best results are **bolded**; second-best is underlined.

Model	Contribution		Like		LongDwell	
	NE	AUC	NE	AUC	NE	AUC
Ordinal RoPE (production)	<u>0.6206</u>	<u>0.9102</u>	<u>0.5985</u>	<u>0.9238</u>	0.8362	0.7597
Timestamp-as-Feature	0.6218	0.9098	0.5997	0.9233	0.8350	0.7603
TO-RoPE	0.6218	0.9095	0.5999	0.9231	<u>0.8349</u>	<u>0.7613</u>
SIREN-RoPE (ours)	0.6182	0.9115	0.5963	0.9249	0.8334	0.7633

train with Adam optimizer and learning rate 10^{-3} with cosine annealing, batch size 800. SIREN-RoPE uses a 2-layer SIREN branch and a 2-layer DNN branch (each 64 hidden units), adding $\sim 0.2\%$ extra parameters over ordinal RoPE.

Evaluation metrics. We report Normalized Entropy (NE, lower is better) and AUC (higher is better) for each of three engagement tasks, defined as follows:

- **Contribution:** a binary label that equals 1 if any engagement action—Like, Share, Comment, Message, or Save—is observed on the post, and 0 otherwise. It captures the broadest notion of active engagement.
- **Like:** a binary label for the explicit Like action alone.
- **LongDwell:** a binary dwell-time engagement label derived from a post-type-specific threshold calibrated on historical data (e.g., ≥ 28 s for video posts, ≥ 15 s for text posts). It captures passive but sustained attention.

NE measures calibration quality of the predicted engagement probability; AUC measures ranking quality. Reporting both dimensions is essential because a model can improve ranking (AUC) without improving calibration (NE), or vice versa.

4.2 Main Results

Table 2 reports held-out engagement prediction performance. Because all models share the same backbone, features, and training procedure (Table 1), every difference in the table is attributable solely to how—or whether—the timestamp is used.

Two findings stand out before examining SIREN-RoPE. First, **Timestamp-as-Feature**—which appends the timestamp to item sequence features while leaving the rotation angle purely ordinal—performs no better than Ordinal RoPE on any metric, and is slightly worse on calibration (NE). This mirrors the ablation finding (Section 4) that temporal signals routed through the embedding dimension fail to help and can interfere with content semantics. Second, **TO-RoPE**—which applies RoPE-style inverse frequencies to the timestamp rather than ordinal position—only slightly improves over Ordinal RoPE on LongDwell task, confirming that adding static time frequencies in a fixed frequency schedule only helps on a very specific task.

SIREN-RoPE, by contrast, improves over Ordinal RoPE consistently across all six metric–task combinations: it reduces Contribution NE by 0.0024 (0.6206 $\rightarrow$ 0.6182), improves Like AUC by 0.0011 (0.9238 $\rightarrow$ 0.9249), and achieves the largest gain on LongDwell AUC (+0.0036, 0.7597 $\rightarrow$ 0.7633), consistent with the hypothesis that engagement is temporally structured and therefore benefits from a rich rotation dimension.

4.3 Ablation Study

Table 3 presents a systematic ablation across six variants, grouped into three questions about the rotation dimension.

(1) Cyclical features improve calibration. Replacing the scalar `time_in_year` with the full five-dimensional ($\cos$, $\sin$) decomposition consistently improves NE. The ($\cos$, $\sin$) representation

Table 3: Ablation study on the production social feed dataset across all six metric–task combinations. NE: lower is better; AUC: higher is better. Best per column in **bold**; second-best underlined. Groups separated by horizontal rules.

Variant	Contribution		Like		LongDwell	
	NE	AUC	NE	AUC	NE	AUC
Semantic rotation (in/out-network)	0.6192	0.9099	0.5972	0.9234	0.8356	0.7604
SIREN-RoPE + <code>time_in_year</code> only	0.6194	<u>0.9115</u>	0.5973	<u>0.9250</u>	0.8350	0.7629
DNN-only + <code>time_in_year</code> only	0.6191	0.9118	0.5972	0.9252	0.8346	<u>0.7630</u>
SIREN-only + full temporal ($f_{\text{DNN}} = 0$)	0.6251	0.9095	0.6029	0.9232	0.8409	0.7600
DNN-only + full temporal ($f_{\text{sin}} = 0$)	0.6180	0.9111	0.5961	0.9244	0.8327	0.7627
Full SIREN-RoPE (ours)	<u>0.6182</u>	<u>0.9115</u>	<u>0.5963</u>	0.9249	<u>0.8334</u>	0.7633

maps each cycle onto a unit circle, eliminating the discontinuous jump (e.g., 86 399 s back to 0 at midnight) that a raw scalar imposes on the network.

(2) SIREN branch: useful for undiscovered periodicities. When cyclical features are already provided, DNN-only with full temporal inputs achieves the best NE, since the $(\cos, \sin)$ pairs already supply the periodic structure SIRENs are designed to learn. We retain the SIREN branch not for the known daily/weekly cycles, but for its capacity to *autonomously discover hidden periodicities* absent from the manual feature decomposition.

(3) Semantic rotation: static signals belong in embeddings. Replacing the temporal input with a binary in/out-of-network feature performs no better than ordinal RoPE. A static social-graph property does not change across time or position, so the dynamic rotation dimension offers no advantage over the embedding space for encoding it. The rotation dimension is most productive with signals that are inherently *relational and dynamic*.

Efficiency analysis. SIREN-RoPE adds a negligible overhead in training ($\sim 1.4\%$ wall-clock time increase over ordinal RoPE at batch size 800, sequence length 1024) and inference ($\sim 1.8\%$). The extra parameters constitute 0.2% of total model parameters. This confirms that the performance improvements come at minimal computational cost.

5 Discussion

Rotation as an expressive dimension. Our results suggest the rotation manifold is a structured expressive channel rather than a mere positional scaffold. Much like the imaginary axis in algebra, a learnable rotation dimension provides an orthogonal space to encode relationships invisible to the embedding space alone. Table 3 confirms this: while injecting temporal features into embeddings is ineffective or detrimental, routing them through the rotation manifold yields consistent gains. This identifies rotation as a qualitatively distinct representation mode, suggesting a unified algebraic structure where semantic (embedding) and dynamic (rotation) dimensions provide non-redundant information.

Complementarity and Generalization. The stability of the learnable gate λ demonstrates that ordinal recency and absolute temporal signals are complementary. This framework extends beyond recommendation to general sequence modeling; for instance, in NLP, the rotary angle could be conditioned on syntactic depth or part-of-speech tags. Such "content-aware" positional geometry allows the attention mechanism to modulate similarity based on structural or semantic context rather than simple proximity, offering a promising direction for more flexible transformer architectures.

6 Conclusion and Future Work

We presented **SIREN-RoPE**, which populates the RoPE rotation manifold with temporal signals via a dual-branch SIREN–DNN network fused with ordinal position through a learnable gate. The broader

argument is that the rotation space is an underexplored second dimension of attention expressivity—the dynamic (imaginary) complement to the semantic (real) embedding space. Experiments on a production social feed dataset confirm consistent gains across calibration and ranking objectives at negligible computational cost, and ablations confirm that the rotation dimension carries signal complementary to and independent of the embedding space.

Future work. Open directions include: (1) a theoretical characterization of what function classes the rotation manifold can express; (2) conditioning f_ϕ on token-type or categorical features for semantic-temporal rotation and potential application in LLM; (3) adapting the framework to cross-attention with separate angle functions per side; and (4) unifying positional encoding across modalities via a shared learnable rotation space.

6.1 Broader Impact Statement

The proposed SIREN-RoPE framework introduces a more expressive method for encoding temporal and semantic signals in Transformer architectures. While primarily a fundamental improvement in representation learning, its applications—particularly in large-scale recommendation and ranking systems—carry significant social implications.

Positive impacts. By enabling models to more accurately capture complex periodicities and user behavior patterns, SIREN-RoPE can lead to more relevant and efficient content discovery. The unified framework may reduce the need for manual, feature-specific engineering, potentially lowering the barrier to developing sophisticated sequential models.

Potential negative impacts. As with any advancement in ranking and recommendation, improved model expressivity could inadvertently reinforce filter bubbles or algorithmic biases if the underlying semantic signals contain historical prejudices.

6.2 Limitations

Extrapolation constraints. The “content-aware” nature of SIREN-RoPE may be constrained by the distribution of temporal and semantic features seen during training. Generalization to out-of-distribution time intervals or entirely new semantic categories remains an open question requiring further empirical study.

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A Attention Modulation of Ordinal RoPE

Figure 2 visualizes what ordinal RoPE actually computes in a recommendation context. We fix a 512-dimensional unit vector as both query and key, place the query at position 0, and sweep the key over ordinal positions 0–1023 for four commonly used bases. Two properties are immediately apparent.

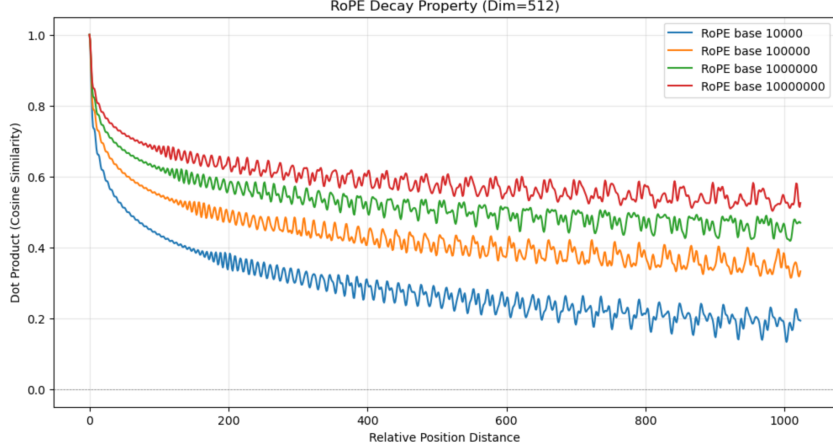


Figure 2: Attention score between a query at position 0 and keys at ordinal positions 0–1023, computed with a 512-dimensional unit vector ($\mathbf{q} = \mathbf{k} = \mathbf{1}/\sqrt{512}$) under four RoPE bases (10^4 , 10^5 , 10^6 , 10^7). All bases produce a monotone global decay with small high-frequency oscillations whose period grows with the base.

Global recency decay. Regardless of base, the attention score decays monotonically with ordinal distance. This is the primary inductive bias ordinal RoPE injects into recommendation models: recent items receive higher attention weight, acting as a soft recency prior.

High-frequency oscillations without semantic meaning. Superimposed on the decay are sinusoidal oscillations whose frequency is determined by the base. In natural language, these oscillations encode syntactic and semantic distance structure. In recommendation, however, ordinal position has no intrinsic periodic meaning—items are indexed by interaction order, not by any cyclical process—so the oscillation pattern carries no useful signal. Instead it acts as structured noise injected into the attention logits.

Ordinal RoPE as a regularizer. This structured noise turns out to be beneficial in a different way: it stabilizes training. In our experiments, replacing no-RoPE attention with ordinal RoPE allowed the learning rate to be increased from 10^{-4} to 10^{-3} without training instability or loss divergence. We therefore attribute two roles to ordinal RoPE in recommendation: (1) a *recency prior* encoded in the global decay, and (2) an *implicit regularizer* encoded in the oscillatory noise. SIREN-RoPE preserves both roles—through the learnable ordinal gate λ —while additionally populating the rotation manifold with genuine temporal semantics.

Base sensitivity in production. To understand whether the choice of base matters for recommendation quality, we swept all four bases (10^4 , 10^5 , 10^6 , 10^7) on our production model and found the downstream NE and AUC metrics to be nearly insensitive to this choice. The differences across bases are within noise, consistent with the observation in Figure 2 that all bases produce the same qualitative decay structure. We selected base = 10^6 for our production deployment based on a negligible but consistent edge in held-out metrics.

B SIREN-RoPE Temporal Attention Sweep

To visualize what temporal structure SIREN-RoPE has actually learned, we extract the trained weights from the production SIREN-RoPE module and build a standalone inference unit that accepts a query timestamp, a key timestamp, and 128-dimensional query/key unit vectors, and returns the rotary-modulated attention score. We fix the query at a reference point and sweep the key timestamp across four spans—one day, one week, one month, and one year—overlapping two consecutive periods in each panel of Figure 3 to expose repeating structure.

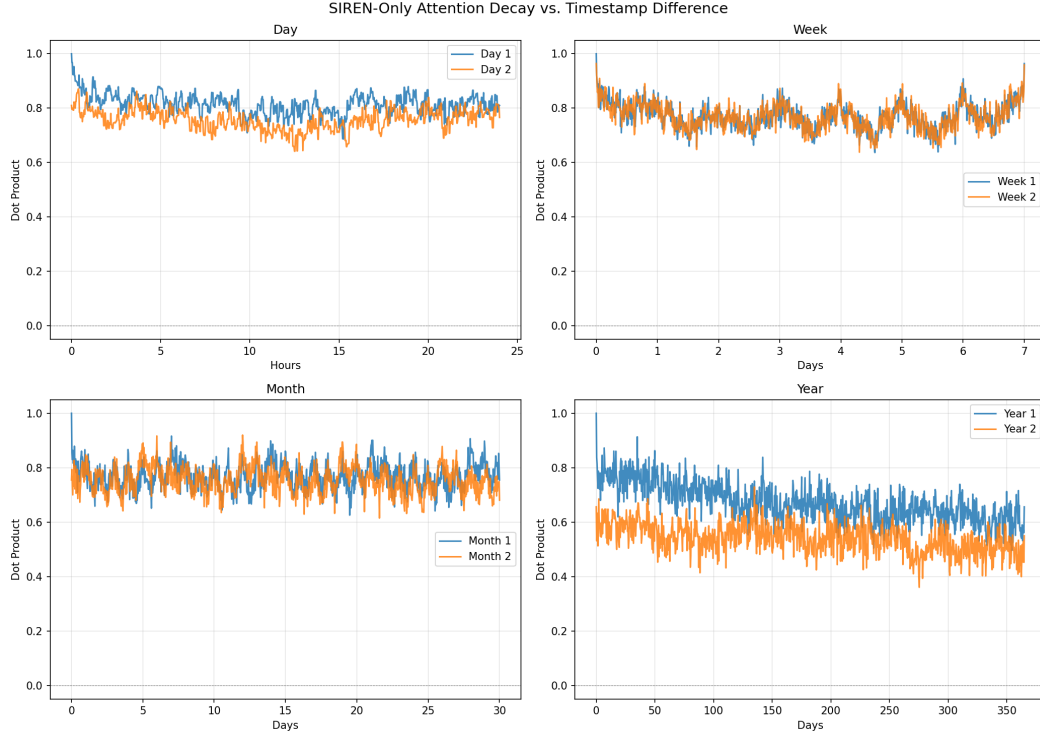


Figure 3: Attention score of a SIREN-RoPE module (weights extracted from the production model) as the key timestamp is swept across four time spans, using a 128-dimensional unit vector for both query and key. Each panel overlaps two consecutive periods (day 1 vs. day 2; week 1 vs. week 2; month 1 vs. month 2; year 1 vs. year 2) to reveal repeating structure.

Strong intra-day periodicity. The day-over-day panel shows that the attention curves for day 1 and day 2 align closely, confirming that the SIREN branch has autonomously recovered a sub-daily (hourly) oscillation pattern that repeats every 24 hours—directly from the training signal, without any explicit hour-of-day feature.

Strong weekly periodicity. The week-over-week panel shows near-perfect overlap of attention scores between week 1 and week 2. The model captures the weekday/weekend behavioral split as a recurring weekly rhythm encoded in the rotation angle. This weekly structure is the most dominant periodicity the network discovers, consistent with pronounced weekday/weekend engagement patterns in social feed data.

No monthly (30 days) periodicity. The month-over-month panel shows no alignment between month 1 and month 2: the curves diverge immediately and do not repeat. This is expected—the dominant weekly period does not divide evenly into 30 days, so no recurring monthly pattern emerges.

Yearly decay only. The year-over-year panel shows a smooth, monotone decay in attention score as the key recedes further into the past, with no recurring annual pattern. This long-range behavior is captured by the aperiodic DNN branch rather than the SIREN branch, confirming the intended division of labor: SIREN encodes high-frequency periodicities (daily, weekly) while the DNN captures low-frequency monotone trends (recency decay).

Frequency-domain analysis. To quantify the learned periodicities, we apply the Fast Fourier Transform (FFT) to the year-long attention curve; Figure 4 shows the result.

The FFT spectrum reveals two dominant peaks: one at ≈ 0.143 cycles/day ($= 1/7 \text{ day}^{-1}$, the weekly frequency) and one at exactly 1 cycle/day (the daily frequency). Both are recovered without any explicit period specification—the model learned these rhythms purely from the training signal.

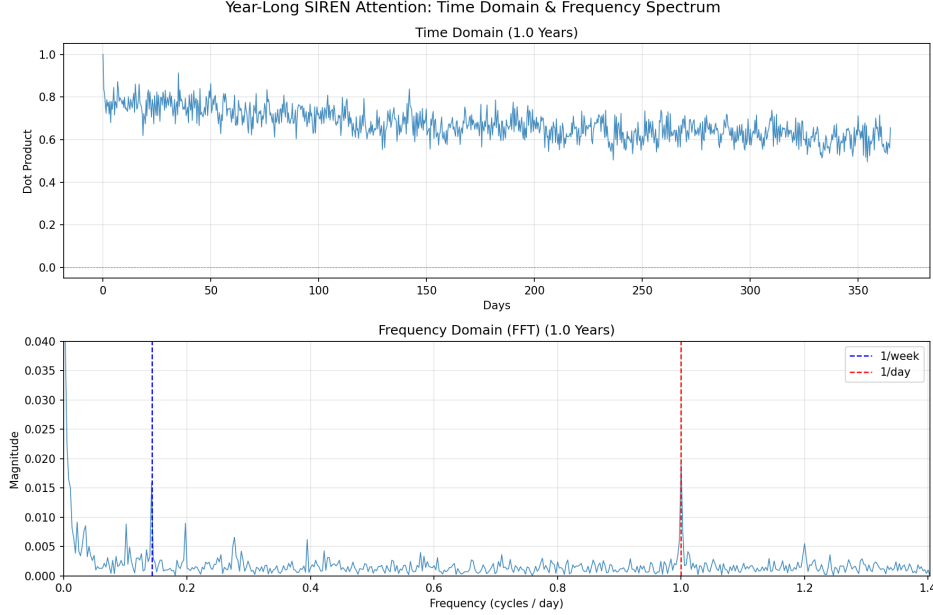


Figure 4: Year-long SIREN-RoPE attention in the time domain (top) and its FFT magnitude spectrum (bottom). Vertical dashed lines mark the weekly ($\frac{1}{7}$ cycles/day) and daily (1 cycle/day) frequencies. Both frequencies exhibit prominent peaks; sub-harmonics at $\frac{1}{2}$, $\frac{1}{3}$, $\dots$ cycles/day are also visible.

These are consistent with common social-feed behavioral rhythms such as weekend-only posting or mid-week engagement surges, and are naturally captured by the harmonic richness of the SIREN branch.

Joint ordinal–temporal attention landscape. Figure 5 provides a unified view by sweeping *both* dimensions simultaneously: the ordinal position of the key (y-axis, 0–120) and the key timestamp (x-axis) across four time spans (1 day, 1 week, 30 days, 365 days).

Two orthogonal patterns are immediately apparent in the heatmap.

Ordinal axis: pure monotone decay. Across all four panels, attention decreases smoothly from ordinal position 0 (most recent) to 120 (oldest) with no oscillatory structure. This confirms that the ordinal component $p_i \cdot \theta_j \cdot \lambda$ contributes a clean recency prior and nothing else—its role is solely to weight recent items more heavily, not to encode any periodic information.

Temporal axis: oscillation reflecting human behavior. In the 1-day and 1-week panels the heatmap exhibits vivid vertical banding: bright and dark stripes repeat with a period of roughly 24 hours (intra-day engagement cycles) and 7 days (weekday/weekend rhythm). The oscillation is consistent across all ordinal positions, confirming it originates entirely from the temporal (SIREN) component $f_\phi(T_i)_j \cdot \omega_j^s$ rather than from the ordinal component. In the 30-day panel the daily banding remains visible but the weekly banding broadens; in the 365-day panel both fine-grained oscillations fade out and the global temporal decay from the DNN branch becomes visible.

Taken together, the time-domain sweeps, frequency-domain analysis, and joint heatmap confirm that SIREN-RoPE learns a multi-resolution temporal geometry directly in the rotation manifold—sharp daily and weekly oscillations reflecting real user behavioral cycles, harmonic sub-peaks at integer-day periods, and a smooth recency decay at the scale of months and years—all without manual specification of these time scales except the feature engineering.

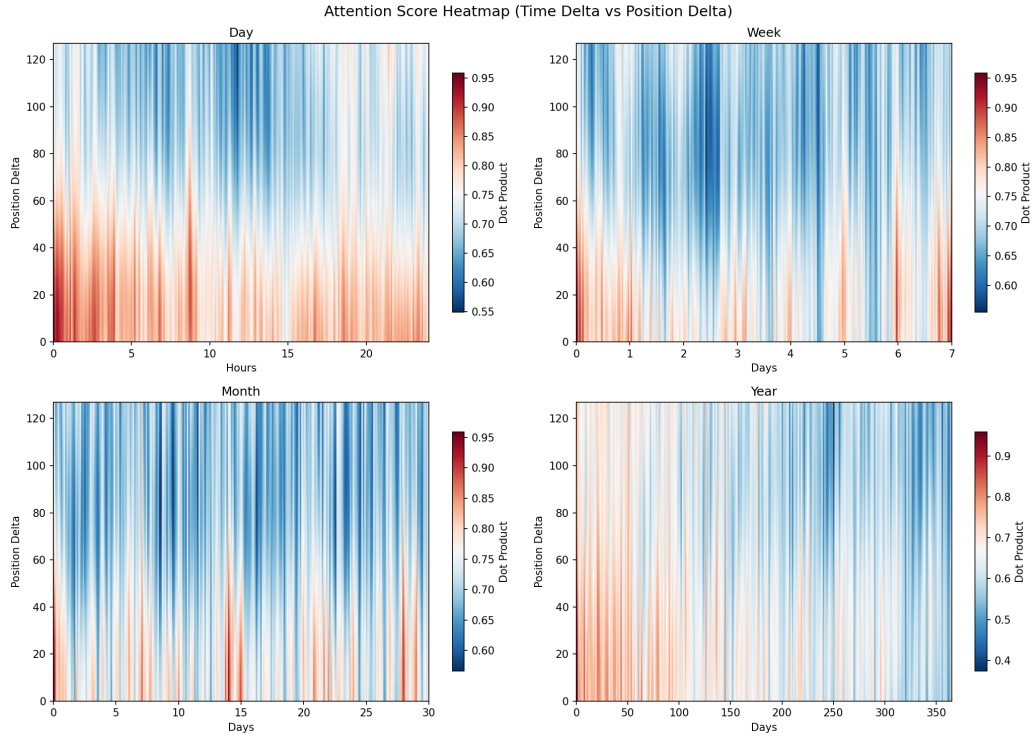


Figure 5: 2D attention score heatmap as a function of key ordinal position (y-axis, 0–120) and key timestamp (x-axis) for four time spans. Color encodes attention magnitude (brighter = higher). The ordinal axis exhibits monotone decay irrespective of the time span; the temporal axis exhibits clear intra-day and intra-week oscillations in the short-span panels.